Technical Note

Power Electronics and Control Systems

Three-phase grid PLLs under grid faults

dq-PLL, DDSRF-PLL and FHT-PLL of the model `frt_dqpll`: continuous-time models, performance on a VDE-AR-N 4110 fault, criticalities and improvements

The three phase locked loops that the model ***`three_phase_dqPLLs_analysis`*** *of* ***`pwr-control/modelization-and-control`*** *runs side by side on the same grid voltage, described from the control point of view: the loop and its front ends, their transfer functions, what the simulated asymmetric fault and a set of test signals show, where each loop fails and what to change, with the DDSRF-PLL first.*

Authors:	**Revision:**	**Date:**
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Abstract: The model `frt_dqpll` feeds the LCL filter of a 250 kW active front end from a 690 V grid emulator and runs three phase locked loops on the capacitor voltage: the synchronous-frame dq-PLL, the decoupled double synchronous reference frame PLL (DDSRF-PLL) and a PLL fed by the positive sequence that two first harmonic trackers build from the $\alpha\beta$ voltages (FHT-PLL). This note writes the three loops as the blocks compute them, derives their continuous-time models (the second order phase loop with its gains, the harmonic observer of the tracker and its positive-sequence calculator, the decoupling network of the DDSRF with its poles) and compares them on the logged simulation of test 25.4 of VDE-AR-N 4110 (phase U to 0.25 pu, V and W to 0.875 pu with a 21.8° shift, 915 ms) and on six test signals, with an offline re-implementation that reproduces the logged FHT-PLL within 4×10^{-5} rad. The dq-PLL needs its gain scheduling in the fault and still shows a 8.6° peak-to-peak ripple at 100 Hz; the FHT-PLL rejects the negative sequence to 1.4° but its phase leads the voltage by 4.15° because of two details of the C code; the DDSRF-PLL as built is fed with the negative-sequence outputs and never locks on the phase, and with the positive-sequence feed alone it locks on $-\omega$ when started from zero frequency. With the positive feed, the frequency integrator started at 1 pu and clamped, and the PI retuned for the low-pass filters in its loop ($k_p = 0.32$, $k_i = 8$), the DDSRF-PLL becomes the best of the three on every test except the frequency step: 0.05° ripple, 2.5° transient at the fault start, 1.3° at its end.



Revision history

Rev.	Date	Description	Authors
00	November 2026	First issue: models of the three PLLs, comparison on the logged fault and on test signals, criticalities and improvements, scripts `py/make_figures.py` and `plotting_results_plls.m`.	Davide Bagnara

Project Repository: modelization-and-control.git (three_phase_dqPLLs_analysis)

Framework: MATLAB/Simulink, Python



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Nomenclature

Symbol	Meaning
PLL	phase locked loop: the estimate of the angle θ and of the pulsation ω of the grid voltage
dq-PLL, SRF-PLL	PLL in the synchronous reference frame (the $\xi\eta$ frame of the model)
DDSRF-PLL	PLL on the decoupled double synchronous reference frame (positive and negative frame)
FHT, FHT-PLL	first harmonic tracker (library C code) and the PLL fed by the positive sequence built from two trackers
abc, $\alpha\beta$, $\xi\eta$	phase quantities, stationary two-axis frame (amplitude-invariant Clarke transform), rotating frame
pos, neg	positive and negative sequence; the positive frame turns by $+\theta$, the negative frame by $-\theta$
pu	per unit of the peak phase voltage of the converter class (690 V: 563 V peak)
θ_g, θ	angle of the grid voltage (positive sequence) and its estimate by the PLL
ω_n, f_0, ω_0	nominal grid pulsation $2\pi f_0$, $f_0 = 50\,\mathrm{Hz}$; ω_0 is the same number as the resonance of the trackers
ω_pu	estimated grid pulsation in pu of ω_n
t_s	control period, 125 µs (4 kHz PWM, double update)
u, $\hat{u}$, $\hat{v}$	grid voltage (pu), its estimate by a tracker and the quadrature estimate ($\hat{v} = \hat{\hat{u}}/\omega_0$)
u_sr	output of the positive-sequence calculator of the FHT-PLL (minus the positive sequence)
u_pos, u_neg	decoupled positive and negative sequence of the DDSRF block, as $\xi\eta$ pairs
V_norm	normalization of the phase detector, $1/\sqrt{\max(\alpha^2+\beta^2,\,0.02)}$
e	input of the loop PI: the normalized η component of the positive sequence (the phase error for small errors)
k_p, k_i	proportional and integral gain of the loop PI (`pll_p`, `pll_i1` of `ctrl_afe_setup`)
ω_c, ζ	natural frequency and damping of the phase loop; ω_x its crossover, PM its phase margin
$T(s)$, $L(s)$	closed loop (phase from the detector input) and open loop of the phase loop
H_1, H_2, D	transfer functions of the tracker, $\hat{u}/u$ and $\hat{\hat{u}}/u$, and their denominator
l_1, l_2, δ	observer gains and damping of the tracker; $L_\mathrm{d} = [l_1\ l_2]$ the discrete gains of the C code



Symbol	Meaning
$G(\omega)$	complex gain of a positive-sequence calculator on a component $e^{j\omega t}$ of $\alpha + j\beta$ ($\omega < 0$: negative sequence)
$F(s)$, ω_f, τ	low-pass filter of the DDSRF, $\omega_\mathrm{f}/(s+\omega_\mathrm{f})$, $\omega_\mathrm{f} = 1/\tau$
A_d, L_dso	matrices of the double-integrator observer of stage B
sat, wrap	saturation between two limits; angle wrapped to $(-\pi, \pi]$: $-\pi + \mathrm{mod}\,(u + 3\pi,\, 2\pi)$
FRT, LVRT	fault ride-through, low-voltage ride-through; `frt_active` is the flag of the FRT detector
z	variable of the z transform, $z = e^{j\omega t_\mathrm{s}}$ on the unit circle

1 Introduction

The model `frt_dqpll.slx` of `foundations_of_power_electronics/three_phase_dqPLLs_analysis` [1] is a test bench for grid synchronization: a 1 MW, 690 V grid emulator with its Dyn11 transformer feeds the LCL filter of a 250 kW active front end whose converter side is open, and the AFE control (125 µs) runs four phase locked loops on the normalized capacitor voltage: the dq-PLL in Simulink, the same loop as a C caller, the DDSRF-PLL and the FHT-PLL. One of them (by default the FHT-PLL) gives the angle and the frequency to the grid-support blocks of the library `afe_uvw_ctrl` [2, 3]. The grid emulator applies test 25.4 of VDE-AR-N 4110 [9] in its asymmetric variant: phase U drops to 0.25 pu, phases V and W to 0.875 pu with a shift of 21.8° each, a type D dip with a positive sequence of 0.625 pu and a negative sequence of 0.375 pu, from 0.75 s for 915 ms.

This note describes the three Simulink loops (the C caller is a copy of the dq-PLL and is not treated) from the control point of view:

- Section 2 writes the loop that the three PLLs share (stage B of the blocks: PI, phase integrator, double integrator observer), its continuous-time model and its discrete poles;
- Section 3 writes the three front ends (stage A): the synchronous frame of the dq-PLL, the first harmonic tracker and the positive-sequence calculator of the FHT-PLL, the decoupling network of the DDSRF-PLL, each with its transfer functions and with the implementation details of the blocks and of the C code that change them;
- Section 4 compares the loops on the logged simulation of the fault and Section 5 on six test signals (phase jump, frequency step, balanced sag, unbalance, harmonics, DC offset);
- Section 6 lists the criticalities of each loop and what to change, with the DDSRF-PLL first, and Section 7 gives the recommendation.

How the results were obtained. The simulation logs the outputs of the FHT-PLL only (the Variant Source removes the other branches from the compiled model), so the comparison runs the three algorithms offline on the logged capacitor voltage `ugd_abc_grid_sim`, step by step as the blocks and the C code compute them: control period, one-step delays, clamps, resets. The offline FHT-PLL reproduces the logged phase `phase_grid_fhtsr_dqpll_sim` within 3.9×10^{-5}

rad (1×10^{-5} rad after the first 20 ms) and the logged tracker signals within 1×10^{-6} pu, which validates the common parts (Clarke transform, normalization, stage B, observer) and the tracker; the dq-PLL and the DDSRF-PLL are transcribed from the diagrams with the same elements. Two scripts do this: `py/make_figures.py` with `py/pll_models.py` in this folder (Python, every figure and number of the note) and `plotting_results_plls.m` in the model folder (MATLAB, the same runs and figures after a simulation, Section 8). The reference angle is the angle of the logged voltage itself: a linear fit of the $\alpha\beta$ angle before the fault, which gives 49.990 Hz (the emulator runs 0.02 % below nominal, the logged FHT-PLL frequency says the same) and holds through the fault because a type D dip does not move the positive sequence (one-period DFT: 0.6254 pu at 0.00° during the fault).

2 The loop shared by the three PLLs

Every PLL of the model is split in two subsystems (Figure 1): stage A turns the three phase voltages into the $\xi\eta$ components of the positive sequence in the frame of the estimated angle, and stage B closes the loop on the η component. Stage B is the same block in the three loops, with the gains of the class `ctrl_afe_setup`: `pll_p` = 1, `pll_i1` = 80 (the DDSRF and FHT loops copy them as `pll_p_ddsrt`, `pll_p_fht`, ...).

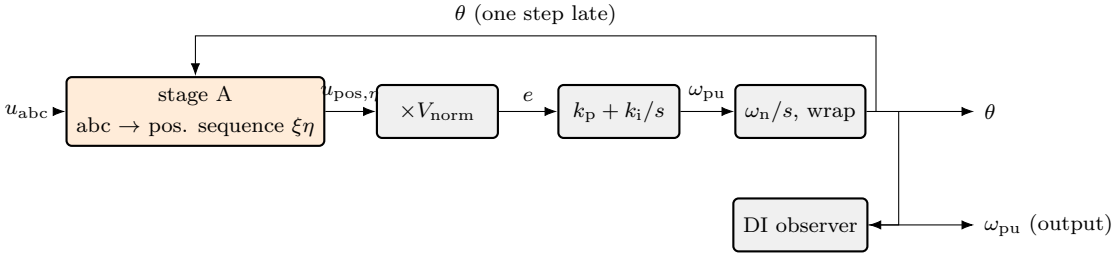


Figure 1: Structure of the three PLLs: stage A (the front end, different in the three blocks) and stage B (the same loop). The normalization V_{norm} is computed in stage A from the $\alpha\beta$ voltage that reaches the rotation.

2.1 Stage B as the block computes it

With η the second output of stage A, $V_{\mathrm{norm}} = 1/\sqrt{\max(\alpha^2 + \beta^2,\, 0.02)}$ and $s_{\mathrm{B}} = \pm 1$ the sign in front of the PI (+1 in the dq-PLL, −1 in the FHT-PLL and in the DDSRF-PLL, where a Gain block of −1 sits before the PI), step k of stage B is

$$e[k] = s_{\mathrm{B}}\, \eta[k]\, V_{\mathrm{norm}}[k], \tag{2.1}$$

$$i[k] = \mathrm{sat}\left(i[k-1] + k_{\mathrm{i}} t_{\mathrm{s}}\, e[k]\right), \qquad \omega_{\mathrm{pu}}[k] = k_{\mathrm{p}}\, e[k] + i[k], \tag{2.2}$$

$$\theta[k] = \mathrm{wrap}\left(\theta[k-1] + t_{\mathrm{s}}\, \omega_{\mathrm{n}}\, \omega_{\mathrm{pu}}[k]\right), \tag{2.3}$$

and the outputs of the block at step k are $\theta[k-1]$ (a Unit Delay) and the frequency estimate of the double integrator observer. The saturation of the integrator is ±2 pu in the dq-PLL and absent in the other two. The integrator starts at zero: every PLL of the model starts from zero



frequency. The angle that stage A uses at step k is the same $\theta[k-1]$, so in steady state, with a constant grid frequency, the output of the block at step k is the angle of the voltage sampled at step k: the one-step delay of the structure costs no phase (the PI drives η to zero on the delayed angle, and the integrator holds the frequency).

The double integrator observer estimates the frequency from the wrapped phase with the states $x = [\hat{\theta},\ \hat{\omega}]^T$:

$$x[k+1] = A_\mathrm{d}\, x[k] + L_\mathrm{dso}\ \mathrm{wrap}\left(\theta[k-1] - \mathrm{wrap}(\hat{\theta}[k])\right), \qquad A_\mathrm{d} = \begin{bmatrix} 1 & t_\mathrm{s} \\ 0 & 1 \end{bmatrix}, \tag{2.4}$$

with the frequency output $\omega_{\mathrm{pu,out}}[k] = \hat{\omega}[k]/\omega_\mathrm{n}$ and $L_\mathrm{dso} = [0.1839,\ 44.78]^T$ from Ackermann's formula at the poles $e^{-\omega_\mathrm{n} t_\mathrm{s}}$ and $e^{-4\omega_\mathrm{n} t_\mathrm{s}}$ (continuous poles $-\omega_\mathrm{n}$ and $-4\omega_\mathrm{n}$, 3.2 ms and 0.8 ms). The frequency output therefore follows the loop frequency ω_pu of (2.2) with about 4 ms of delay and without its proportional kick; it is this output that the rest of the control reads (`omega_grid_pll_pu`) and that the figures of this note show.

2.2 Continuous-time model and gains

For small errors $\eta\, V_\mathrm{norm} \approx \sin(\theta_\mathrm{g} - \theta) \approx \theta_\mathrm{g} - \theta$ (the normalization removes the voltage amplitude), and (2.2)–(2.3) are the second order loop

$$\dot{\theta} = \omega_\mathrm{n}\Big(k_\mathrm{p}\, e + k_\mathrm{i} \int e\Big), \quad e = \theta_\mathrm{g} - \theta + d \qquad \Longrightarrow \qquad T(s) = \frac{\theta}{\theta_\mathrm{g} + d} = \frac{\omega_\mathrm{n}(k_\mathrm{p} s + k_\mathrm{i})}{s^2 + \omega_\mathrm{n} k_\mathrm{p}\, s + \omega_\mathrm{n} k_\mathrm{i}}, \tag{2.5}$$

where d collects everything that enters the detector besides the phase error: the negative sequence at 2ω, the harmonics 6ω, 12ω, ..., a DC offset at ω. The natural frequency is $\omega_\mathrm{c} = \sqrt{\omega_\mathrm{n} k_\mathrm{i}}$ and the damping $\zeta = \omega_\mathrm{n} k_\mathrm{p}/(2\omega_\mathrm{c})$. The loop is of type 2 for the angle: zero steady-state error for a frequency step. Table 1 gives the numbers of the gain sets that appear in this note; the open loop $L(s) = \omega_\mathrm{n}(k_\mathrm{p} s + k_\mathrm{i})/s^2$ crosses at $\omega_\mathrm{x} \approx \omega_\mathrm{n} k_\mathrm{p}$ ($L \approx \omega_\mathrm{n} k_\mathrm{p}/s$ well above the PI corner $k_\mathrm{i}/k_\mathrm{p}$), so with $k_\mathrm{p} = 1$ the loop crosses at 324 rad/s (52 Hz) and attenuates the 100 Hz term of the detector only to $|T(j2\omega_\mathrm{n})| = 0.475$ (Figure 2, left). The last two columns anticipate Section 3.3: with the low-pass filter $F(s) = \omega_\mathrm{f}/(s + \omega_\mathrm{f})$ of the DDSRF in the loop, the same gains lose half of their phase margin.

Table 1: Gain sets of the phase loop (2.5): the gains of the model, the fault gains of the dq-PLL (`pll_p_frt`), the alternative fault set of the README of the model and the set proposed for the DDSRF-PLL. Crossover ω_x and phase margin without and with the low-pass filter $\omega_\mathrm{f} = \omega_\mathrm{n}$ of the DDSRF in the loop.

gain set	k_p	k_i	ω_c [rad/s] (Hz)	ζ	$\lvert T(j2\omega_\mathrm{n})\rvert$	$\lvert T(j6\omega_\mathrm{n})\rvert$	without LPF		with LPF	
							ω_x [rad/s]	PM	ω_x [rad/s]	PM
model	1	80	159 (25.2)	0.99	0.475	0.166	324	76°	255	33°
dq-PLL in the fault	0.2	80	159 (25.2)	0.20	0.126	0.034	165	22°	155	−5°
alternative fault set	0.2	3.2	32 (5.0)	0.99	0.100	0.033	65	76°	64	64°
proposed, DDSRF-PLL	0.32	8	50 (8.0)	1.00	0.159	0.053	103	76°	99	58°

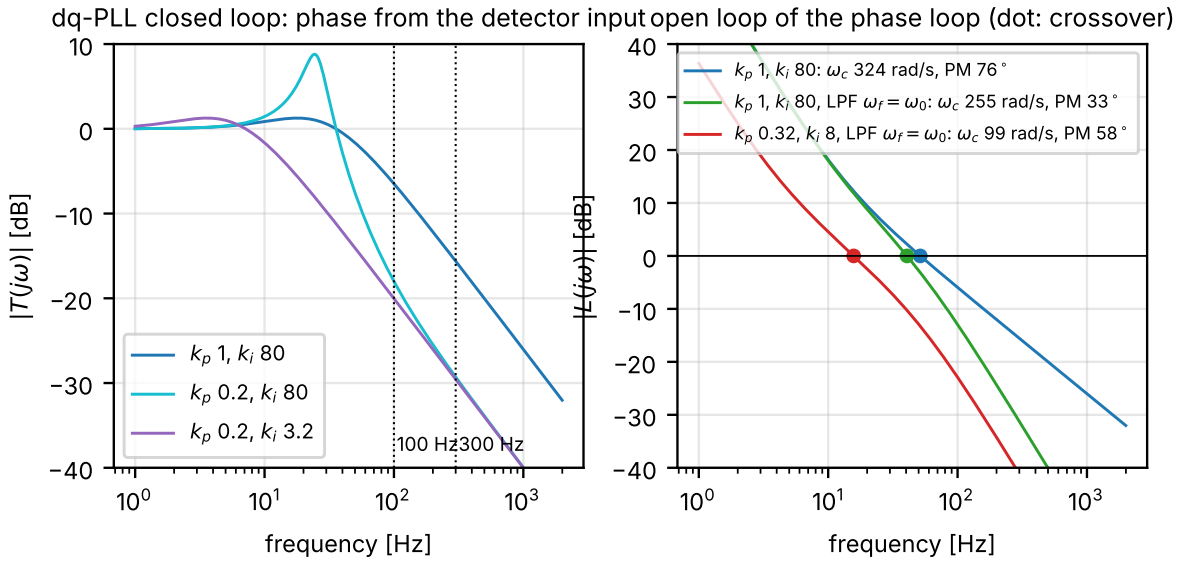


Figure 2: Left: closed loop $|T(j\omega)|$ of the phase loop for the three gain sets of the dq-PLL; the dotted lines mark the 100 Hz of the negative sequence and the 300 Hz of the 5th and 7th harmonics. Right: open loop of the same PI with and without the low-pass filter of the DDSRF-PLL in the loop; the dots mark the crossover.

Discrete loop. With the forward Euler integrators of (2.2)–(2.3) and the one-step delay of the angle, the poles of the discrete loop are $z = 0.9794,\ 0.9809$ for the normal gains, i.e. $s = -166,\ -154$ rad/s against the continuous $-157 \pm 21j$: at 125 µs the discretization changes nothing that matters, for any of the gain sets (`pll_design` of the model init prints the same numbers). The fault gains give the poles $-31 \pm 156j$ rad/s, a damping of 0.20: a phase jump rings at 25 Hz for about 150 ms.

3 The three front ends

3.1 dq-PLL: the synchronous frame

Stage A of the dq-PLL is the Clarke transform (amplitude invariant, $\alpha = \frac{2}{3}(u_a - \frac{1}{2}u_b - \frac{1}{2}u_c)$, $\beta = (u_b - u_c)/\sqrt{3}$), the normalization and the rotation by $-\theta$ (chart `alphabeta2xieta_pos`):

$$\xi = \alpha\cos\theta + \beta\sin\theta, \qquad \eta = -\alpha\sin\theta + \beta\cos\theta, \tag{3.1}$$

so that for $\alpha + j\beta = Ve^{j\theta_g}$ the detector gives $\eta\, V_{norm} = \sin(\theta_g - \theta)$. Nothing filters the voltage: a negative sequence $V_n e^{-j\theta_g}$ gives $\eta \ni -V_n \sin(\theta_g + \theta) \approx -V_n \sin 2\theta_g$, a term at 2ω that the loop attenuates only by $|T(j2\omega_n)|$; the 5th and 7th harmonics give a term at 6ω, a DC offset of the measurement a term at ω. The normalization uses the total $\alpha\beta$ magnitude $|V_p + V_n e^{-j2\theta_g}|$, which during an unbalance oscillates at 2ω too and modulates the loop gain between $1/(V_p + V_n)$ and $1/(V_p - V_n)$.

The only countermeasure of the block is the scheduled proportional gain: the Switch of `scheduled-p-gain` takes k_p = `pll_p_frt` = 0.2 while the FRT detector of the library

(`frt_vde4110`, on the averaged sequences) holds `frt_active`, and keeps $k_\mathrm{i} = 80$. This reduces the ripple gain from 0.475 to 0.126 (Table 1) at the price of the damping: ζ drops from 0.99 to 0.20, and the recovery of the normal gain at the end of the fault happens with the integrator loaded by the ripple.

3.2 FHT-PLL: first harmonic trackers and the positive-sequence calculator

The tracker. The first harmonic tracker of the library (`first_harmonic_tracker.c`, parameters `fht_grid` of `ctrl_afe_setup`) is a Luenberger observer of a damped harmonic oscillator at $\omega_0 = \omega_\mathrm{n}$, with the state x_1 the estimate $\hat{u}$ of its input and x_2 the estimate of its derivative:

$$\dot{x}_1 = x_2 + l_1(u - x_1), \qquad \dot{x}_2 = -\omega_0^2\, x_1 - \delta\omega_0\, x_2 + l_2(u - x_1), \qquad \hat{u} = x_1, \quad \hat{v} = x_2/\omega_0, \tag{3.2}$$

with $\delta = 0.05$ and the gains placed at the poles $-\omega_0$ and $-4\omega_0$: the characteristic polynomial $s^2+(l_1+\delta\omega_0)s+l_1\delta\omega_0+\omega_0^2+l_2 = (s+\omega_0)(s+4\omega_0)$ gives $l_1 = 4.95\,\omega_0 = 1555/\mathrm{s}$ and $l_2 = 2.7525\,\omega_0^2$. The transfer functions from the input are

$$H_1(s) = \frac{\hat{u}}{u} = \frac{l_1 s + l_1\delta\omega_0 + l_2}{D(s)}, \qquad H_2(s) = \frac{\hat{\dot{u}}}{u} = \frac{l_2 s - l_1\omega_0^2}{D(s)},$$
$$D(s) = s^2 + (l_1 + \delta\omega_0)s + l_1\delta\omega_0 + \omega_0^2 + l_2. \tag{3.3}$$

At $s = j\omega_0$ the term ω_0^2 of D cancels s^2 and $H_1(j\omega_0) = 0.993\angle -0.25°$: a unity-gain band-pass whose width is set by the observer poles, $l_1 \approx 5\omega_0$, i.e. about 250 Hz. This is the point to keep in mind: the tracker converges in 4 ms (5 % error after 3.9 ms from a cold start) because it is wide, and for the same reason it passes the 5th and 7th harmonics with a gain of 0.35 and 0.40 and a DC offset with 0.72 (Table 2). It is a fast estimator of the fundamental, not a selective filter.

The positive-sequence calculator. Stage A of the FHT-PLL (Figure 3) runs one tracker on α and one on β and combines the in-phase and quadrature estimates as

$$u_{\mathrm{sr},\alpha} = -\tfrac{1}{2}\big(\hat{u}_\alpha + \hat{v}_\beta\big), \qquad u_{\mathrm{sr},\beta} = -\tfrac{1}{2}\big(\hat{u}_\beta - \hat{v}_\alpha\big), \qquad \text{i.e.} \qquad u_\mathrm{sr} = -\tfrac{1}{2}\big(\hat{u} - j\hat{v}\big) \ \text{ in complex form,} \tag{3.4}$$

which is the classic positive-sequence calculator built on a quadrature signal generator: for a positive sequence $e^{j\omega_0 t}$ the quadrature is $\hat{v} = j\hat{u}$ and (3.4) gives $-\hat{u}$, for a negative sequence $e^{-j\omega_0 t}$ it is $\hat{v} = -j\hat{u}$ and the output is zero. The minus sign is compensated by the Gain -1 in front of the PI of stage B ($s_\mathrm{B} = -1$), and the normalization uses $|u_\mathrm{sr}|$, which is the positive sequence only. For a component $e^{j\omega t}$ of $\alpha + j\beta$ (with $\omega < 0$ for a negative sequence) the output is $-G(\omega)\, e^{j\omega t}$ with

$$G(\omega) = \tfrac{1}{2}\Big(H_1(j\omega) - j\,\frac{H_2(j\omega)}{\omega_0}\Big),$$
$$G(\omega_0) = \frac{(l_1\delta\omega_0 + 2l_2) + 2j\,l_1\omega_0}{2D(j\omega_0)} = 0.982\angle +0.80°, \qquad |G(-\omega_0)| = \frac{l_1\delta\omega_0}{2|D(j\omega_0)|} = 0.021. \tag{3.5}$$

The damping δ is the only reason why the calculator is not exact at ω_0: with $\delta = 0$ the quadrature state is the exact derivative in steady state, $G(\omega_0) = 1$ and $G(-\omega_0) = 0$. With $\delta = 0.05$ the positive sequence leads the input by 0.8° and 2.1 % of the negative sequence leaks into the output.

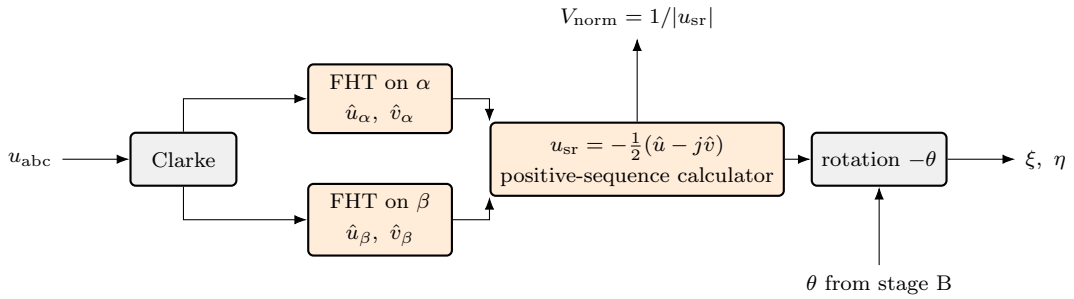


Figure 3: Stage A of the FHT-PLL. The two trackers are two instances of the library C code with the parameters `fht_grid` (resonance $\omega_0 = \omega_{\mathrm{n}}$, $\delta = 0.05$, gains L_{d}), held in reset for the first 12.5 ms.

The discrete implementation. The C code integrates (3.2) with the second order Taylor matrix of the oscillator ($A_{\mathrm{d}} = I + At_{\mathrm{s}} + \frac{1}{2}A^2t_{\mathrm{s}}^2$, Section 7 of the README of the model) and with the gains $L_{\mathrm{d}} = [0.1819,\ 29.59]$ designed with Ackermann's formula on the forward Euler matrix $I + At_{\mathrm{s}}$ at the poles $e^{-\omega_0 t_{\mathrm{s}}}$, $e^{-4\omega_0 t_{\mathrm{s}}}$. Two details of `first_harmonic_tracker_process` change the response at the fundamental:

1. the innovation is computed with the output stored at the *previous* call, $u[k] - \hat{u}[k-1]$, not with the current predicted state: $x[k+1] = A_{\mathrm{d}}\,x[k] + L_{\mathrm{d}}\,(u[k] - x_1[k-1])$. The observer has three poles instead of two, $z = 0.962,\ 0.805,\ 0.230$ ($s = -312,\ -1733,\ -11\,776$ rad/s) instead of the designed $z = 0.961,\ 0.854$ ($-320,\ -1264$): still stable and fast, but the output $\hat{u}[k] = x_1[k]$ now *leads* a 50 Hz input by 1.96° (gain 0.992) where the designed observer lags by 0.28°;
2. the quadrature output is the state *after* the update, $\hat{v}[k] = x_2[k+1]/\omega_0$, one step ahead of $\hat{u}[k] = x_1[k]$. At 50 Hz one step is 2.25°.

Table 2 and Figure 4 give the resulting gain of the calculator for the components of interest, next to the continuous model and to two variants that change only the code: the same observer with consistent states ($\hat{u} = x_1[k]$, $\hat{v} = x_2[k]/\omega_0$ and the innovation on the current state), and the same with $\delta = 0$ (gains recomputed for the same poles). The leak of the negative sequence doubles in the code as it is (3.9 % instead of 2.1 %: the quadrature of a negative sequence is one step off) and the positive sequence leads the sampled voltage by 4.15°. The lead depends on the frequency, about 0.8° per hertz, because the tracker is tuned at the fixed $\omega_0 = \omega_{\mathrm{n}}$ and does not follow the PLL frequency.

Table 2: Gain $|G|$ and angle of the positive-sequence calculator of the FHT-PLL on a component of $\alpha + j\beta$ at the given frequency (negative frequency: negative sequence). Continuous model (3.5); library C code as it is (delayed innovation, quadrature one step ahead); same observer with consistent states; consistent states and $\delta = 0$.

component	continuous	C code as is	consistent states	consistent, $\delta = 0$
fundamental, +50 Hz	0.982 / 0.8°	0.981 / 4.1°	0.981 / 0.8°	1.000 / 0.0°
negative sequence, −50 Hz	0.021	0.039	0.021	0.000
+49 Hz	0.983 / 1.6°	0.982 / 4.9°	0.982 / 1.7°	1.001 / 0.9°
+51 Hz	0.981 / 0.0°	0.980 / 3.4°	0.980 / 0.0°	0.999 / −0.8°
−51 Hz	0.018	0.034	0.019	0.010
DC	0.724	0.716	0.716	0.722
2nd, +100 Hz	0.856	0.892	0.861	0.881
5th, −250 Hz	0.345	0.350	0.350	0.362
7th, +350 Hz	0.398	0.520	0.405	0.416
11th, −550 Hz	0.219	0.212	0.224	0.231
13th, +650 Hz	0.224	0.325	0.229	0.236

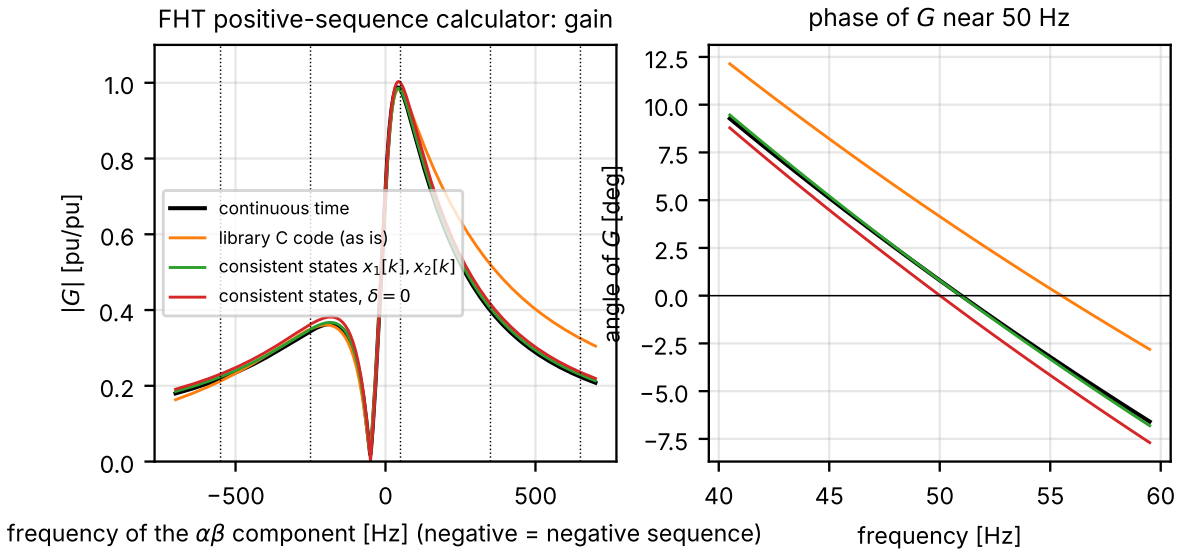


Figure 4: Positive-sequence calculator of the FHT-PLL: gain on a component of $\alpha + j\beta$ against its (signed) frequency, dotted lines at the harmonics of interest (left), and the phase of the output around the fundamental (right), for the continuous model, the C code as it is and the two variants of Table 2.

3.3 DDSRF-PLL: the decoupling network

Stage A of the DDSRF-PLL (Figure 6) pre-filters each phase voltage with a tracker (three instances of the C code, output $\hat{u}$ only), takes the Clarke transform of the three estimates, computes V_{norm} from it and feeds the block `sequences/ddsrf`: the rotations by $-\theta$ (positive frame, (3.1)) and by $+\theta$ (negative frame, chart `alphabeta2xieta_neg`: $\xi = \alpha\cos\theta - \beta\sin\theta$,



$\eta = \alpha \sin\theta + \beta \cos\theta$), four library low-pass filters with $\tau =$ `tau_ddsrf` $= 1/\omega_n$ and the cross decoupling of [7]. In complex form, with $u = \alpha + j\beta$, $p = u_{pos,\xi} + j u_{pos,\eta}$, $n = u_{neg,\xi} + j u_{neg,\eta}$ and $F(s) = \omega_f/(s + \omega_f)$, $\omega_f = 1/\tau$:

$$p = F(s)\Big[u\, e^{-j\theta} - n\, e^{-j2\theta}\Big], \qquad n = F(s)\Big[u\, e^{+j\theta} - p\, e^{+j2\theta}\Big]. \tag{3.6}$$

In the positive frame the positive sequence is a constant and the negative sequence turns at -2ω; the network subtracts from each frame the *filtered* estimate of the other sequence rotated into it. The two properties of (3.6) that the comparison needs are:

- **the steady state is exact.** With $\theta = \omega t$, $u = V_p e^{j\omega t} + V_n e^{-j\omega t}$, writing $p = P_0 + P_2 e^{-j2\omega t}$, $n = N_0 + N_2 e^{j2\omega t}$ and using $F(0) = 1$ gives $P_0 = V_p - N_2$, $N_2 = F(j2\omega)(V_p - P_0)$, hence $P_0 = V_p$, $P_2 = 0$, $N_0 = V_n$, $N_2 = 0$: the decoupled outputs carry no 2ω term at all, whatever ω_f. This is why the ripple of the DDSRF-PLL in Sections 4 and 5 is two orders of magnitude below the one of the dq-PLL, and why ω_f is a choice about dynamics only;
- **the dynamics.** A perturbation $\tilde{p}$ at the complex frequency s in the positive frame reaches the negative frame at $s + j2\omega$ and comes back: $\tilde{p} = F(s)\, F(s + j2\omega)\, \tilde{p}$, so the sequence estimation error decays with the roots of $1 - F(s)F(s + j2\omega) = 0$,

$$s = -(\omega_f + j\omega) \pm \sqrt{\omega_f^2 - \omega^2}. \tag{3.7}$$

  For $\omega_f \le \omega$ both roots have the real part $-\omega_f$; for $\omega_f > \omega$ they split and one of them slows down towards $-\omega^2/(2\omega_f)$. The fastest decay is at $\omega_f = \omega$, a double root at $-\omega - j\omega$ (3.2 ms): the model's choice $\tau = 1/\omega_n$ is the optimum of (3.7) (Table 3, Figure 5). The price is the attenuation of what the filters are also asked to remove from the frames: the 6th harmonic passes with $|F(j6\omega)| = 0.164$ and a DC offset, which appears at ω in both frames, with 0.707.

Table 3: Roots (3.7) of the decoupling network and attenuation of the low-pass filter for different corner frequencies; $\omega_f = \omega_0$ is the value of the model.

ω_f	τ [ms]	root 1 [rad/s]	root 2 [rad/s]	slowest time constant [ms]	$\lvert F(j2\omega_0)\rvert$	$\lvert F(j6\omega_0)\rvert$
$\omega_0/2$	6.37	$-157 - 42j$	$-157 - 586j$	6.37	0.243	0.083
$\omega_0/\sqrt{2}$	4.50	$-222 - 92j$	$-222 - 536j$	4.50	0.333	0.117
ω_0 (model)	3.18	$-314 - 314j$	$-314 - 314j$	3.18	0.447	0.164
$1.5\,\omega_0$	2.12	$-120 - 314j$	$-822 - 314j$	8.33	0.600	0.243
$2\,\omega_0$	1.59	$-84 - 314j$	$-1172 - 314j$	11.9	0.707	0.316

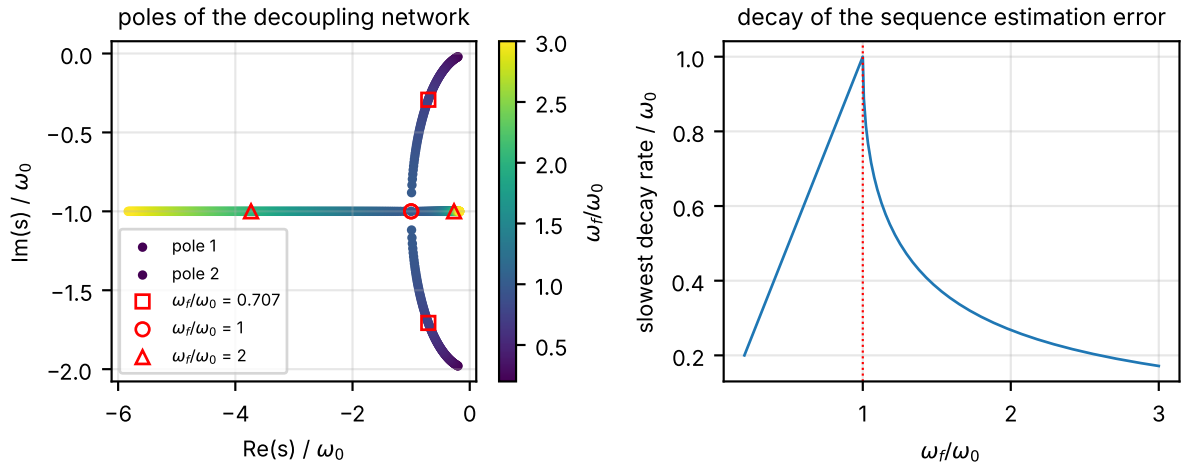


Figure 5: Roots of the decoupling network (3.7) as ω_f/ω_0 goes from 0.2 to 3 (left; the markers are $1/\sqrt{2}$, 1 and 2) and the slowest decay rate (right): the fastest separation of the sequences is at $\omega_f = \omega_0$.

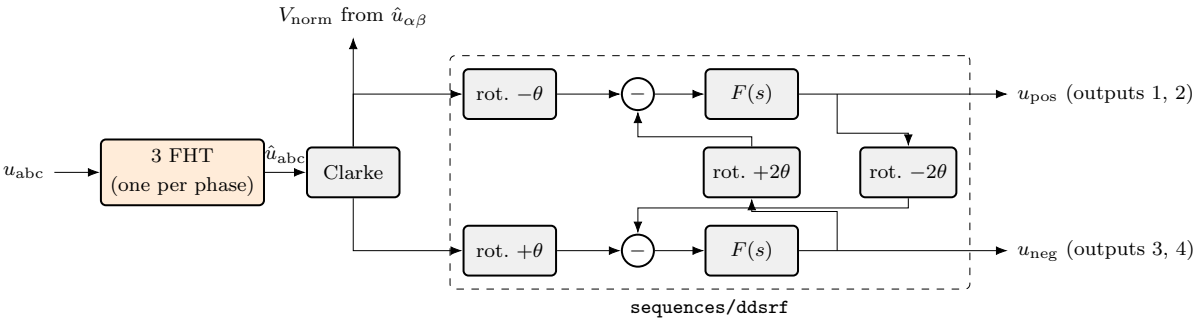


Figure 6: Stage A of the DDSRF-PLL. Stage B of the model is wired to the outputs 3 and 4 (the negative sequence) through the Gain -1; the loop of [7] and of the corrected variants of this note takes the outputs 1 and 2.

The loop with the filter in it. The decoupled η of the positive sequence is $F(s)$ applied to the phase error (for small errors and no unbalance), so the open loop of the DDSRF-PLL is $F(s)\,L(s)$. With the gains of the model the crossover moves from 324 to 255 rad/s and the phase margin from 76° to 33° (Table 1, Figure 2 right): the step responses of Section 5 ring and settle in 165 ms where the dq-PLL settles in 35 ms. The fault gains of the dq-PLL would be unstable here (PM −5°). A PI chosen for this loop is in the last row of the table: $k_p = 0.32$ gives $\omega_x \approx \omega_n k_p = 100$ rad/s (16 Hz), $k_i = 8$ puts the PI corner at 25 rad/s, a quarter of the crossover, and the margin is 58° with the filter.

The wiring of the model. In `frt_dqpll.slx` the outports of the subsystem `sequences` are named `u_grid_pos_xi`, `u_grid_pos_eta` but are connected to the outputs 3 and 4 of the `ddsrf` block, the *negative* sequence, and stage B multiplies them by -1 before the PI. The loop therefore drives $u_{\mathrm{neg},\eta}$ to zero. For a negative sequence $V_n e^{-j(\omega t-\varphi)}$ the negative-frame η is $V_n \sin(\theta - \omega t + \varphi)$, so the loop locks θ on the mirrored angle of the negative sequence, $\omega t - \varphi$:

during the fault it is a negative-sequence PLL. With a balanced grid the negative outputs are zero whatever θ is and the loop has no phase information: the frequency still converges towards ω through the 2ω terms that an angle error leaks into the negative frame before the decoupling settles, but the phase is arbitrary (Section 4: $-40°$ to $-52°$ before the fault, drifting). The intended loop, the one of [7], takes the outputs 1 and 2 with $s_B = +1$; the note calls it "positive feed".

4 Performance on the simulated fault

The logged run is 2.25 s long, with the fault from 0.75 s to 1.665 s (Figure 7): the capacitor voltage shows the 20 ms ramps of the fault generator, the LCL ringing at 1284 Hz at every step (README of the model) and the two sequences of the type D dip, 0.625 and 0.375 pu, with the positive sequence at the same angle as before the fault. The seven variants of Table 4 are run on this voltage; Figures 8 to 11 show them. "Lock" is the last time before the fault at which the phase error (minus its steady-state offset) exceeds 1°; "settle" the same after the fault start and after its end; "ripple" the peak-to-peak phase error between 1.2 and 1.65 s. The flag `frt_active` is not logged: the scheduled gain of the dq-PLL is applied on the fault window itself (the detector of the library reacts within a few milliseconds).

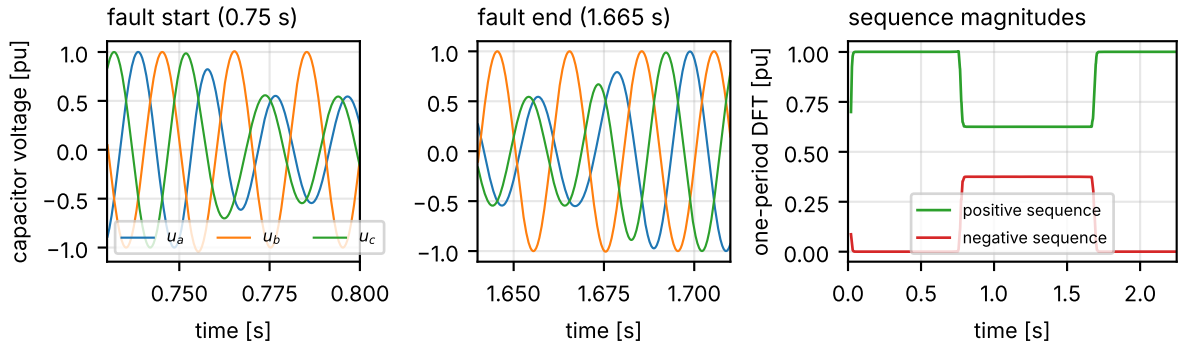


Figure 7: Logged capacitor voltage (pu) at the start and at the end of the fault, and the sequence magnitudes from a one-period DFT on the reference angle.

Table 4: The PLL variants on the logged fault. Offset: mean phase error against the voltage angle before the fault; lock from zero frequency; peak and settling (to 1°) at the fault start and end; peak-to-peak ripple of the phase and of the frequency output inside the fault. The DDSRF-PLL with the positive feed and zero start locks on $-\omega$ (its "ripple" of 360° is the error wrapping). *Improved: positive feed, integrator started at 1 pu and clamped to [0.5, 1.5], $k_p = 0.32$, $k_i = 8$.

variant	offset [deg]	lock [ms]	fault start peak [deg]	settle [ms]	ripple [deg pp]	ω ripple [pu pp]	fault end peak [deg]	settle [ms]
dq-PLL, k_p 1 fixed	0.00	37	19.5	915	33.6	0.460	15.4	21
dq-PLL, scheduled k_p (model)	0.00	37	4.9	915	8.6	0.120	16.2	23
FHT-PLL	4.15	59	1.8	23	1.38	0.019	1.9	16
DDSRF-PLL as built (neg. feed)	−48.7	no	228	no	0.10	0.001	172	no
DDSRF-PLL, pos. feed, zero start	$-\omega$ lock	no	180	no	360	0.001	180	no
DDSRF-PLL, pos. feed, init 1 pu	1.96	158	3.9	26	0.33	0.004	3.5	24
DDSRF-PLL improved*	1.96	107	2.5	36	0.05	0.000	1.3	32

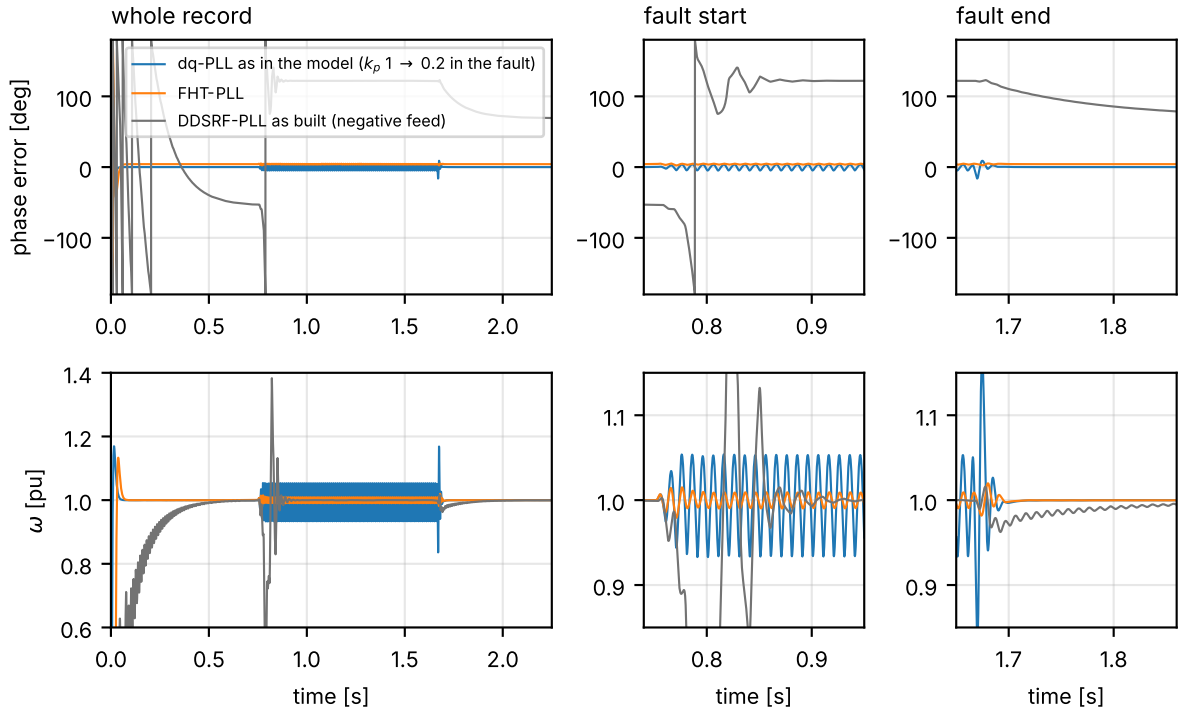


Figure 8: The three PLLs as they are in the model, on the logged voltage: phase error against the voltage angle (top) and frequency output (bottom), whole record and zooms at the fault start and end. The DDSRF-PLL as built never locks on the phase; the dq-PLL carries the 100 Hz ripple through the whole fault.

dq-PLL. With the normal gains the 0.375 pu negative sequence gives a 33.6° peak-to-peak ripple at 100 Hz on the phase and 0.46 pu on the frequency output: linear prediction $2 \cdot (V_n/V_p) \cdot |T(j2\omega_n)| = 2 \cdot 0.6 \cdot 0.475$ rad $= 32.7°$, the rest is the modulation of the normalization. The scheduled gain brings the ripple to 8.6° (prediction 8.7°) and the frequency ripple to 0.12 pu,

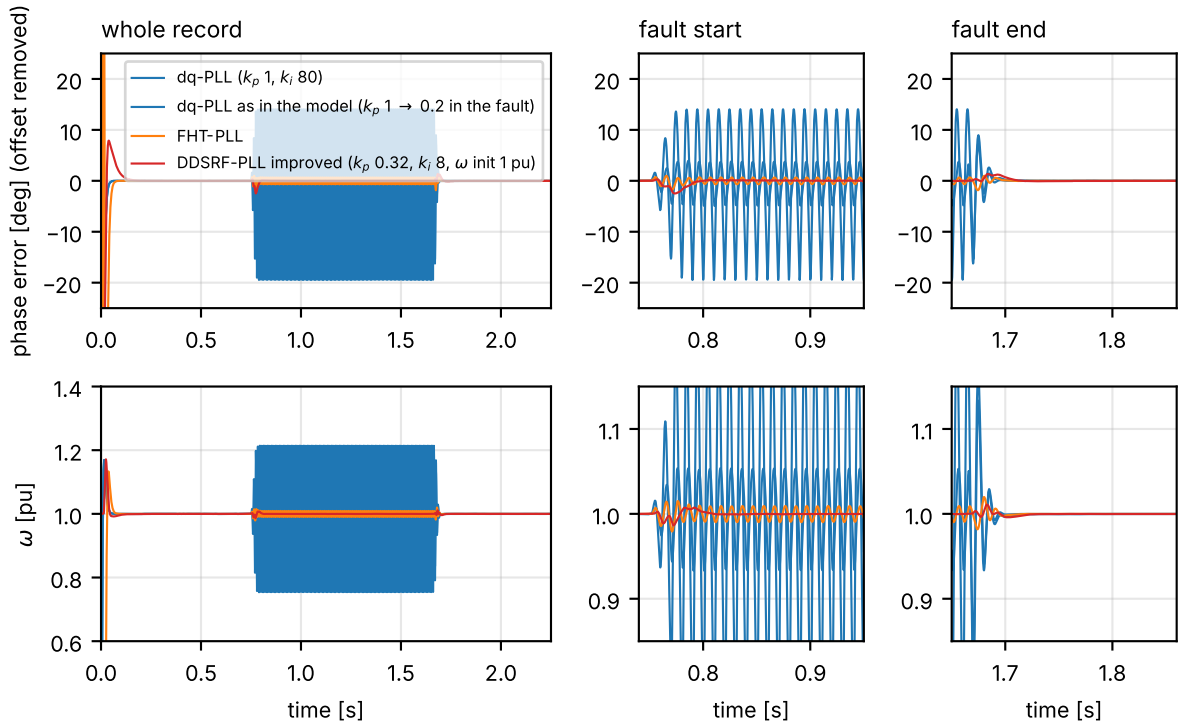


Figure 9: Same run, steady-state offsets removed: the dq-PLL with constant and with scheduled gain, the FHT-PLL and the improved DDSRF-PLL.

still enough to disturb the FRT state machine and the reactive current references that read `omega_grid_pll_pu` and `phase_grid_pll`. The transient at the fault start is small with the fault gains (4.9°) but the recovery at the fault end, when the normal gain comes back with the integrator loaded by the ripple, peaks at 16.2°: the same as without scheduling.

FHT-PLL. Lock in 59 ms from zero frequency (the trackers are held in reset for 12.5 ms), 1.8° at the fault start, 1.9° at its end, 23 and 16 ms to settle, a ripple of 1.38° peak-to-peak: the leak of 3.9 % of Table 2 on 0.375 pu of negative sequence is 0.0147 pu against the 0.613 pu of positive sequence that the calculator outputs, 1.37° peak-to-peak on the angle of u_{sr} (measured on the logged u_{sr}: 2.81°, with the normalization and the DFT bias of a finite record), attenuated by $|T(j2\omega_{\mathrm{n}})| = 0.475$. The phase of the FHT-PLL leads the sampled voltage by 4.15° all the time (4.154° on the logged phase, 4.154° on the logged u_{sr}, 1.958° on the logged tracker outputs: the numbers of Table 2).

DDSRF-PLL as built. The frequency converges (0.9966 pu before the fault, 0.9998 inside it) but the phase does not: −40° at 0.5 s, −52° at 0.7 s, then a lock at +122° inside the fault, which is the mirrored angle of the negative sequence, and a drift after the fault. The block output $|u_{\mathrm{neg}}|$ follows the true negative sequence inside the fault (0.372 pu) because the loop is locked on it. The positive feed alone, started from zero frequency as every PLL of the model, converges to $\omega_{\mathrm{pu}} = -0.9989$ with $|u_{\mathrm{neg}}| = 0.992$ pu: the network explains a positive sequence seen from a frame turning backwards as a negative sequence, with no η error left, so the lock on $-\omega$ is a

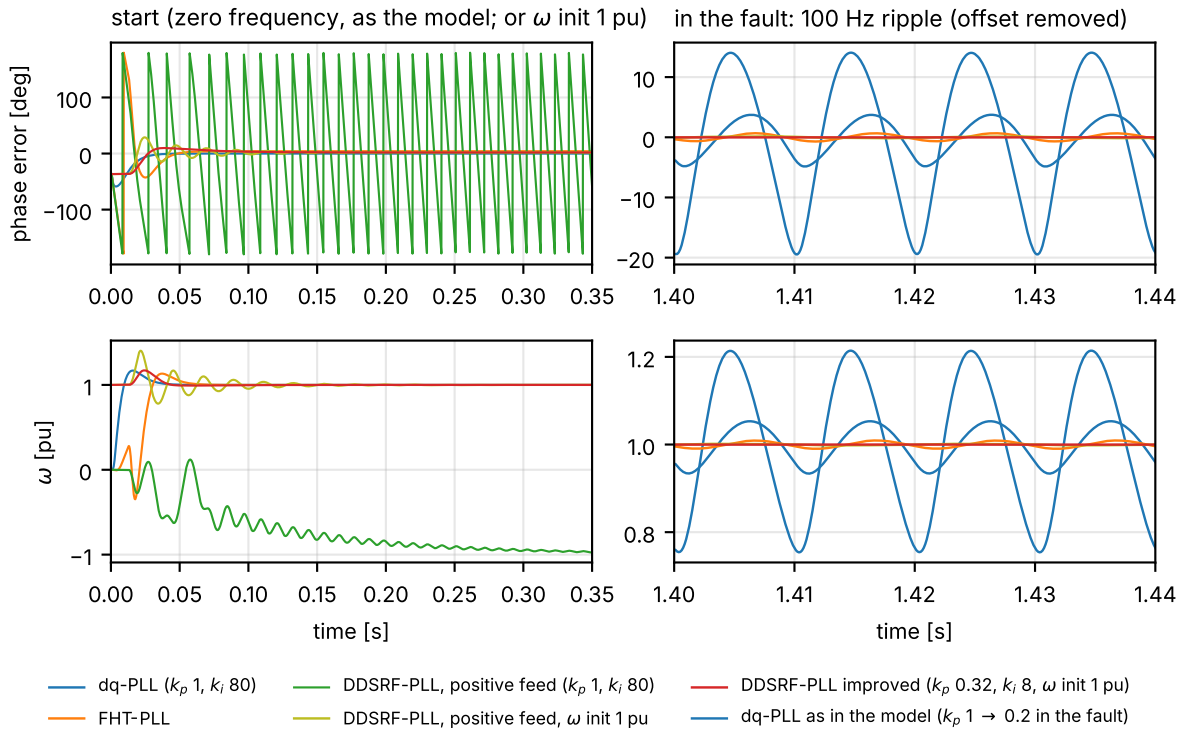


Figure 10: Left: start from zero frequency as in the model (dq-PLL, FHT-PLL, DDSRF-PLL with the positive feed) and with the integrator initialized at 1 pu (two DDSRF variants); the positive feed started from zero converges to $\omega_{\mathrm{pu}} = -1$. Right: 40 ms inside the fault, the 100 Hz ripple of the five variants.

stable equilibrium of the DDSRF-PLL that the dq-PLL and the FHT-PLL do not have (their detector keeps a 2ω term there that pulls the loop out). With the integrator started at 1 pu the positive feed locks in 158 ms (the margin of 33° shows in the slow, ringing acquisition) and goes through the fault with a 3.9° transient and a 0.33° ripple. The improved variant (positive feed, integrator at 1 pu clamped to [0.5, 1.5], $k_{\mathrm{p}} = 0.32$, $k_{\mathrm{i}} = 8$) is the best line of Table 4: 2.5° at the fault start, 1.3° at its end, 0.05° of ripple, 4×10^{-4} pu on the frequency output; its offset of 1.96° is the lead of the three pre-filter trackers (Section 3.2).

5 Performance on test signals

The six tests of Table 5 are ideal three-phase sets at 125 µs, with every loop started locked at 1 pu, the event at 0.4 s and, for the dips, its end at 0.8 s. Five variants are compared: the dq-PLL with the normal gains and with the fault gains held all the time, the FHT-PLL, the DDSRF-PLL with the positive feed and the gains of the model, and the improved DDSRF-PLL. The figures 12 and 13 show the phase error with its steady-state offset removed and the frequency output.

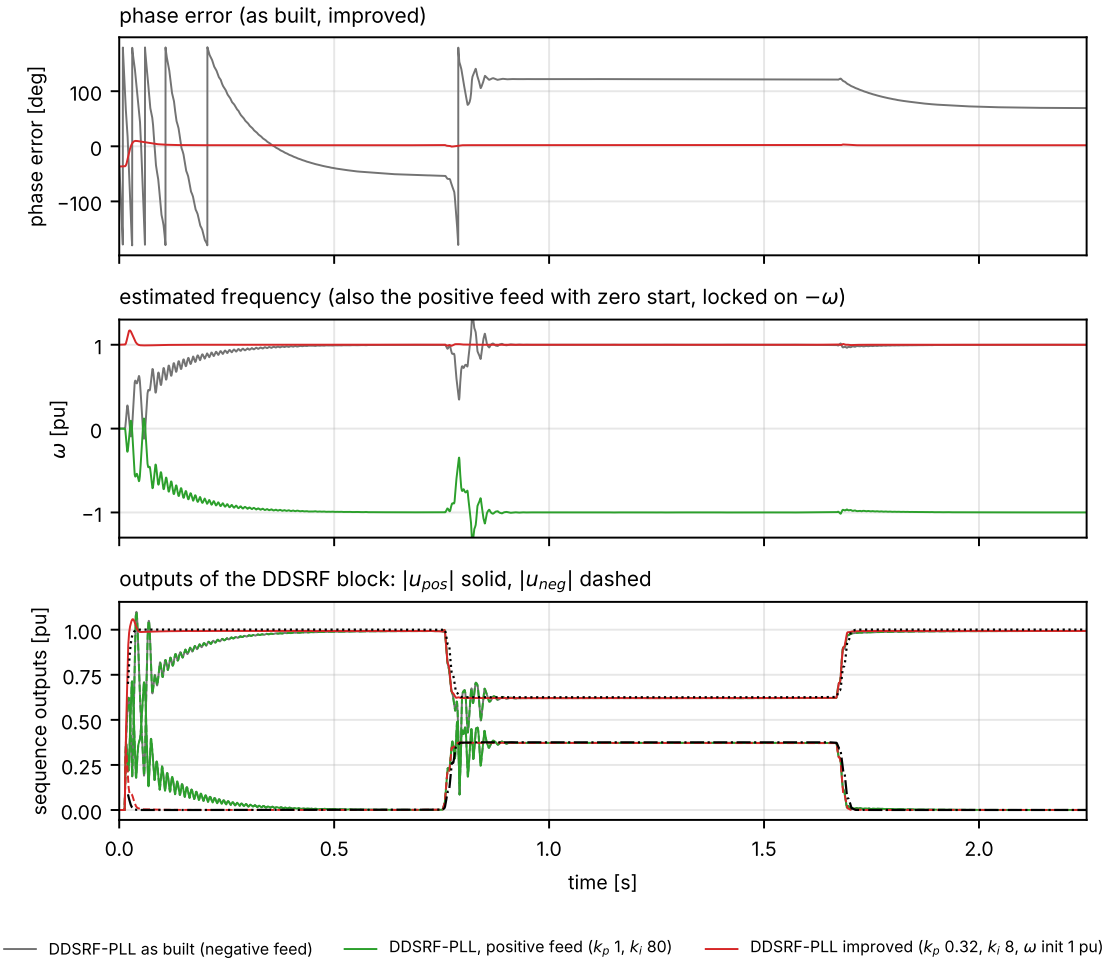


Figure 11: DDSRF-PLL variants on the logged voltage: phase error of the loop as built and of the improved loop (top), frequency output of the three variants (middle), magnitude of the two sequence outputs of the `ddsrf` block against the true sequences (bottom). As built, the block separates the sequences correctly only during the fault, when the loop is locked on the negative sequence.

Table 5: Test signals: peak phase error [deg] / time to settle within 0.5° [ms] / peak-to-peak ripple in steady state [deg]. "400" means not settled within the 400 ms of the event (a permanent ripple). Harmonics: 5th 5 % negative, 7th 3 % positive, 11th 2 % negative, 13th 1.5 % positive. DC offset: 2 % of the peak voltage on phase a.

variant	phase jump 30°	frequency step +1 Hz	sag to 0.3 pu
dq-PLL, k_p 1, k_i 80	30 / 35 / 0	0.84 / 15 / 0	0 / 0 / 0
dq-PLL, fault gains (k_p 0.2)	30 / 124 / 0.01	1.72 / 36 / 0	0 / 0 / 0
FHT-PLL	30 / 37 / 0	1.61 / 400 / 0	8.2 / 17 / 0
DDSRF-PLL, pos. feed, model gains	30 / 165 / 0.13	1.69 / 54 / 0	42.4 / 191 / 0.26
DDSRF-PLL improved	30 / 107 / 0	3.37 / 103 / 0	11.6 / 30 / 0
variant	type D dip	harmonics	DC offset 2 %
dq-PLL, k_p 1, k_i 80	21.4 / 400 / 33.5	0.32 / 0 / 0.39	0.67 / 400 / 1.28
dq-PLL, fault gains (k_p 0.2)	12.8 / 400 / 8.6	0.12 / 0 / 0.08	0.59 / 23 / 0.65
FHT-PLL	9.5 / 400 / 1.34	0.26 / 0 / 0.42	0.49 / 0 / 0.94
DDSRF-PLL, pos. feed, model gains	11.7 / 161 / 0.04	0.18 / 0 / 0.08	2.91 / 400 / 5.1
DDSRF-PLL improved	3.1 / 28 / 0	0.07 / 0 / 0.03	0.37 / 0 / 0.49

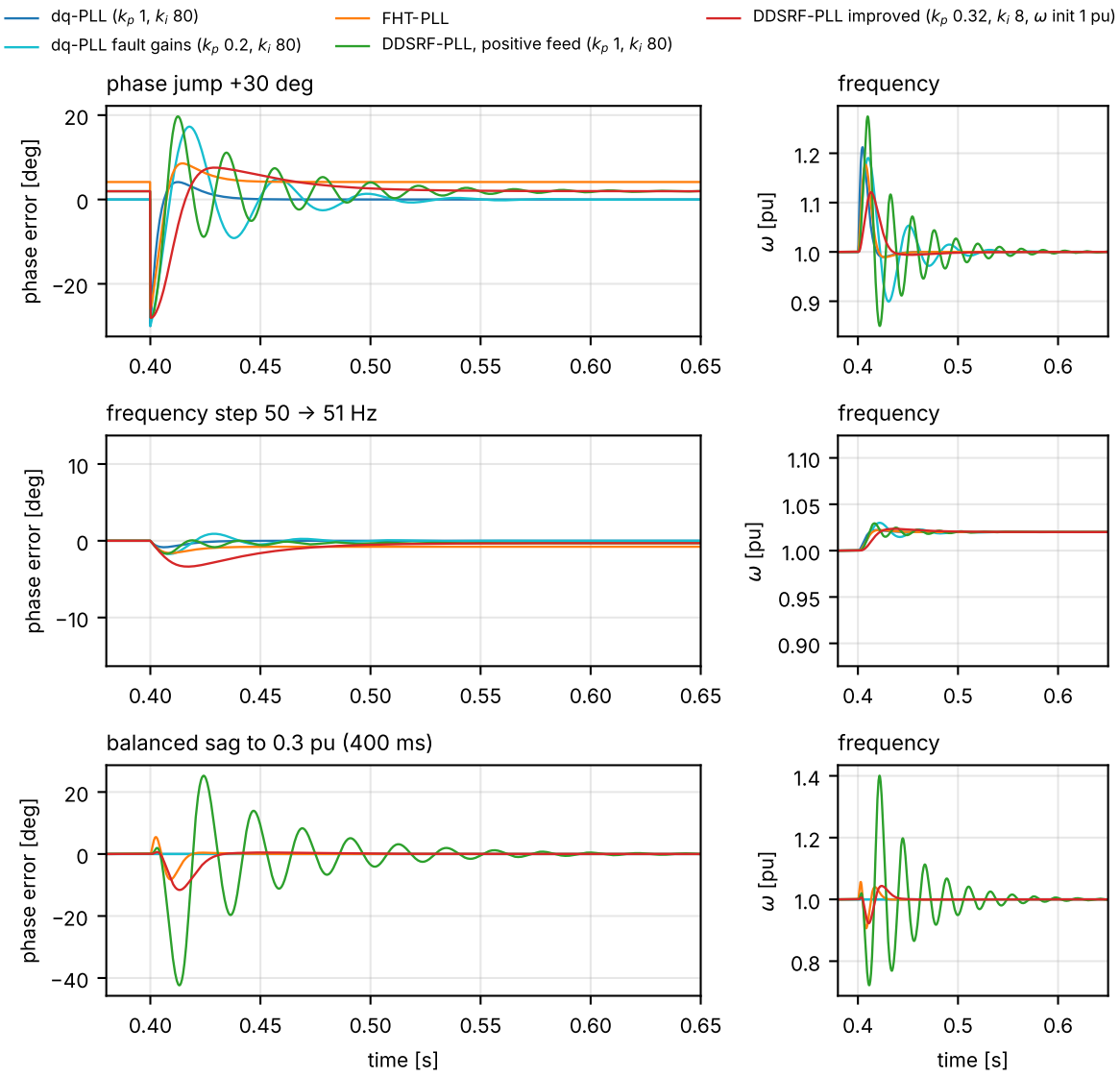


Figure 12: Test signals, steps: phase jump of 30°, frequency step from 50 to 51 Hz, balanced sag to 0.3 pu. Phase error with the steady-state offset removed (left) and frequency output (right).

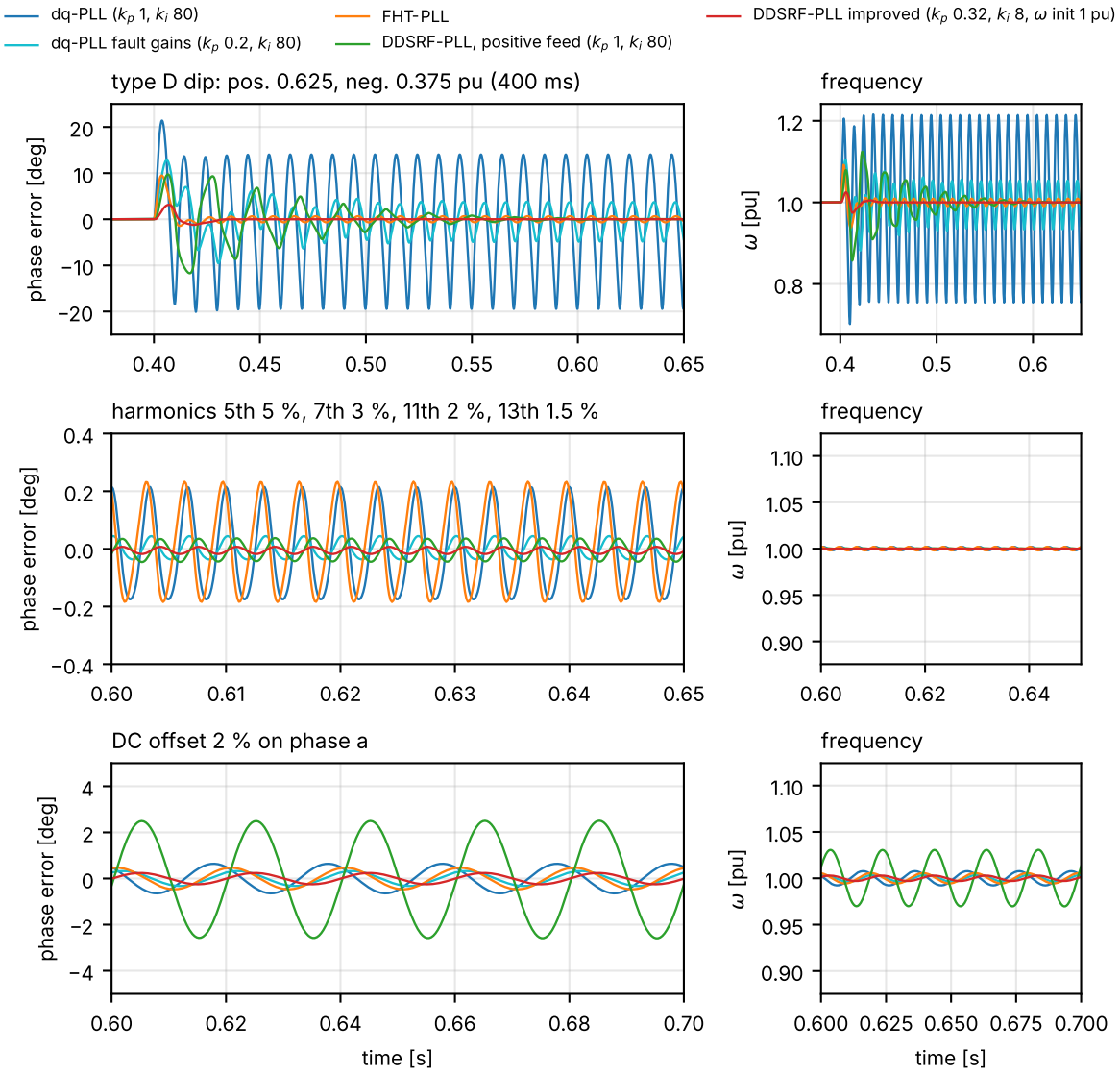


Figure 13: Test signals, disturbances: type D dip (positive 0.625, negative 0.375 pu), harmonics, DC offset of 2 % on phase a. Phase error with the steady-state offset removed (left, zoomed in time for the two steady-state tests) and frequency output (right).

- **Phase jump.** All loops peak at the 30° of the jump; the dq-PLL and the FHT-PLL settle in 35 ms (the damping 0.99 of the loop), the fault gains in 124 ms (damping 0.20), the DDSRF-PLL in 165 ms with the model gains (margin 33°) and in 107 ms with the proposed ones.

- **Frequency step.** A 2 % step gives 0.8° on the dq-PLL and 1.5° to 1.7° on the others; the slower improved DDSRF-PLL peaks at 3.4°. The FHT-PLL never settles within 0.5°: its steady-state lead changes by 0.8° between 50 and 51 Hz (Table 2), a frequency-dependent bias that the type 2 loop cannot remove because it is in the front end.

- **Balanced sag.** The dq-PLL does not see it (the normalization is exact). The FHT-PLL shows the transient of its trackers, 8.2° for 17 ms. The DDSRF-PLL swings to 42° with



the model gains and to 129° without the pre-filters (not in the table): at the step the positive-frame filter still holds the old 1 pu, the decoupling injects the difference, rotated at 2ω, into the negative frame and back into the positive η, and the normalization by the new 0.3 pu amplifies the false error by 3.3. The improved gains bring it to 11.6°, the pre-filters help because they smooth the step. A deeper sag makes this worse: it is the first criticality of the DDSRF-PLL on balanced faults.

- **Type D dip.** The ripple separates the three structures: 33.5° for the dq-PLL (8.6° with the fault gains, the number of the logged run), 1.34° for the FHT-PLL (the leak of its calculator), below 0.05° for the DDSRF-PLL (exact steady state). The transient at the dip is 3.1° for the improved DDSRF-PLL against 9.5° to 21° for the others.

- **Harmonics.** The dq-PLL and the FHT-PLL are the same (0.4° peak-to-peak): the tracker gives no selectivity at the 5th and 7th. The low-pass filters of the DDSRF-PLL add their attenuation (0.08°, 0.03° with the lower gains).

- **DC offset.** 2 % on one phase gives 1.3° on the dq-PLL and 0.9° on the FHT-PLL (the tracker passes DC with 0.72), but 5.1° on the DDSRF-PLL with the model gains: the offset appears at ω in both frames, each filter passes it with 0.707 and the decoupling adds the two contributions. With the lower gains it is 0.49°, the same as the dq-PLL with its fault gains. A real measurement chain has offsets of this order; this is the second criticality of the DDSRF-PLL.

6 Criticalities and what to change

6.1 DDSRF-PLL

1. **Wiring of stage B.** The loop reads the negative-sequence outputs (3 and 4) of the `ddsrf` block through the Gain −1 (Section 3.3): it is a negative-sequence PLL during the fault and has no phase lock otherwise. Connect the outputs 1 and 2 (the outport names of `sequences` already say `u_grid_pos_*`) and remove the Gain −1 (or keep it and swap the sign of the chart, but the two other loops use $s_\mathrm{B} = +1$ on η). Every other item below assumes this change.

2. **Lock on $-\omega$.** The network has a stable equilibrium with the loop turning backwards, and the loop reaches it when it starts from zero frequency (Figure 10). Start the frequency integrator (2.2) at 1 pu and clamp it to a window around nominal, [0.5, 1.5] pu in this note (the dq-PLL block clamps at ±2, which still allows −1). The same initialization removes the 40 to 160 ms start-up transient of every PLL of the model and the risk of a wrong lock after a loss of signal.

3. **Gains.** The filter $F(s)$ is in the loop: with $k_\mathrm{p} = 1$, $k_\mathrm{i} = 80$ the phase margin is 33° (step responses ring, 165 ms to settle) and the fault gains of the dq-PLL would be unstable. The DDSRF-PLL does not need a gain reduction in the fault, because the negative sequence does not reach the detector; it needs its own PI. The proposed $k_\mathrm{p} = 0.32$, $k_\mathrm{i} = 8$ (crossover



100 rad/s, margin 58°, damping 1.0) improves every test but the frequency step: fault transient 3.9° → 2.5°, balanced sag 42° → 11.6°, DC offset 5.1° → 0.49°, settling 165 → 107 ms. A middle set ($k_\mathrm{p} = 0.5$, $k_\mathrm{i} = 20$, tested but not shown) settles in 64 ms with 4.8° at the dip and 1.3° on the DC offset.

4. **Balanced sags.** The false error that the decoupling builds during an amplitude step, amplified by the normalization on the reduced voltage, is structural; the lower gains and the pre-filters reduce it. Two more measures are possible if deep balanced dips matter (VDE-AR-N 4110 asks to ride through 0 pu for 150 ms): a slower or rate-limited normalization (the floor of 0.02 on $\alpha^2 + \beta^2$ lets V_norm reach 7), or a freeze of the loop frequency below a voltage threshold, which the FRT detector already knows.

5. **DC offset.** The two frames add their contributions and the model gains give 5.1° for 2 % of offset. Besides the gains, a third branch of the network that estimates the $\alpha\beta$ offset from the residual $u - pe^{j\theta} - ne^{-j\theta}$ through a slow low-pass ($\omega_\mathrm{f}/4$) and subtracts it from both frames was tried in `pll_models.py` (option `dc_tau`): it removes the 50 Hz ripple completely with the proposed gains (0.00°), but it is unstable with the model gains and costs 5.7° instead of 2.6° at the unbalanced dip and 36° instead of 24° at the sag (variant without pre-filters). A high-pass on the phase voltages before the block is the simpler alternative.

6. **Pre-filters.** The three trackers add the 1.96° lead of Section 3.2 to the angle of the DDSRF-PLL, and they filter little (Table 2); they do smooth the steps (sag 11.6° with them, 24° without, with the proposed gains). Keep them only after the C code is corrected (item 2 of Section 6.2), or accept and compensate the lead.

7. ω_f**.** $\tau = 1/\omega_\mathrm{n}$ is the fastest separation (Table 3); $\omega_\mathrm{f} = \omega_\mathrm{n}/\sqrt{2}$, often quoted, trades 1.3 ms of separation time for 30 % less of the 6th harmonic and of the DC term in the frames. Not a priority once the gains are right.

8. **Normalization.** A normalization by the decoupled positive sequence $|u_\mathrm{pos}|$ instead of the total $|\alpha + j\beta|$ was tried: unstable at the unbalanced dip with the model gains (the loop gain rises while the filtered estimate lags). If it is used, bound it from below.

6.2 FHT-PLL

1. **Phase lead of 4.15°.** The angle that the default PLL of the model gives to the rest of the control leads the sampled voltage by 4.15° at 50 Hz, 4.9° at 49 Hz, 3.4° at 51 Hz (Table 2), while the dq-PLL is aligned to the sample. A 4° error on the frame moves 7 % of the converter current between active and reactive axis and shifts the sequences that `grid_voltage_sequences` and the FRT detector compute. If a lead is wanted as a compensation of the modulator delay ($1.5\,\omega t_\mathrm{s} = 3.4°$ at 50 Hz), it should be explicit, $\theta + k\,\omega_\mathrm{pu}\omega_\mathrm{n}t_\mathrm{s}$, and the same for every PLL.

2. **C code of the tracker.** The innovation uses the output of the previous call and the quadrature the state after the update (Section 3.2). With $u[k] - x_1[k]$ in the innovation



and $\hat{v}[k] = x_2[k]/\omega_0$ the calculator has the gain of the continuous design, $0.981\angle 0.8°$, and the leak of the negative sequence halves (2.1 %, ripple 0.7° instead of 1.4° in the fault). With $\delta = 0$ (the observer gains give all the damping needed; recompute L_d for the same poles) the calculator is exact at ω_0: $G = 1$, no leak, no lead. Both are changes to `first_harmonic_tracker.c` and its setup, used by every model that calls it.

3. **Selectivity.** With the poles at $-\omega_0$, $-4\omega_0$ the tracker is a 250 Hz wide band-pass: harmonics and DC pass with 0.2 to 0.7 and the FHT-PLL is no better than the dq-PLL on them (Table 5). Slower poles (e.g. $-\omega_0/2$, $-\omega_0$) narrow it and slow the start and the fault transients in proportion: a choice to make per application, not a bug.

4. **Fixed resonance.** $\omega_0 = \omega_\mathrm{n}$ does not follow the PLL frequency; the lead of 0.8° per hertz and the leak at off-nominal frequency are the cost. The C code takes ω_0 as an argument at every call: feeding it $\omega_\mathrm{pu}\omega_\mathrm{n}$ (filtered) from stage B makes the calculator adaptive at no cost in the block.

5. **Loop.** No clamp on the integrator (a lost lock winds up, see the DDSRF item 2) and the start from zero frequency; the trackers' reset of 12.5 ms delays the lock to 59 ms.

6.3 dq-PLL

1. **Negative sequence.** The structure has no rejection of the 100 Hz term: 33.6° peak-to-peak with the normal gains on the test 25.4 dip. The scheduled gain is a patch that depends on the FRT detector (a few milliseconds of latency on the averaged sequences, during which the full ripple reaches the frequency output) and still leaves 8.6° and 0.12 pu of frequency ripple.

2. **Fault gains.** $k_\mathrm{p} = 0.2$ with $k_\mathrm{i} = 80$ gives a damping of 0.20: the fault transient rings at 25 Hz (4.9° here, 124 ms to settle on a phase jump). If the aim is a slower loop in the fault, k_i must scale with k_p^2 ($k_\mathrm{i} = 3.2$ keeps 0.99 at 5 Hz); the ripple changes little (6.9° instead of 8.6°) because $|T(j2\omega_\mathrm{n})|$ is set by k_p. The return to the normal gain at the end of the fault, with the integrator loaded by the ripple, gives the 16° recovery peak: release the gain through a ramp, or after the ripple is gone.

3. **Normalization.** The total $\alpha\beta$ magnitude oscillates at 2ω during an unbalance and modulates the loop gain; a normalization by a filtered magnitude (one period moving average) removes the term that the linear prediction does not cover.

4. **Start and clamp.** The ±2 pu clamp of the integrator is the only one of the three loops, but it still allows −1 pu; initialize at 1 pu and clamp around it as for the DDSRF-PLL.

6.4 Common points

- The three loops start from zero frequency and lock in 37 to 160 ms; the initial condition $i[0] = 1$ pu is free.



- The frequency that the control reads comes from the double integrator observer, 4 ms behind the loop; the loop frequency ω_{pu} of (2.2) is available in the block and carries no extra delay (but the proportional term of the ripple).
- The outputs are one step late by construction and the FHT-PLL adds 4.15°: whichever PLL the Variant Source selects, the downstream blocks see a different angle. The selection flag is a design parameter of the model; the angle it gives should not depend on it.
- The reference angle of this note is the sampled voltage; the ADC delay of one step and the quantization of the model are the same for the three loops and are not in the comparison.

7 Summary

Table 6 puts the three structures side by side as they come out of Sections 4 to 6. For the grid-support control of the model, which needs a clean positive-sequence angle and a frequency without 100 Hz ripple during asymmetric faults, the DDSRF-PLL with the positive feed, the initialized and clamped integrator and the lower PI gains is the recommendation: the exact steady-state separation of the sequences is a property that the other two do not have, and its weak points (balanced sags, DC offset, slower frequency tracking) are known and bounded once the gains fit the filter in the loop. The FHT-PLL is a good second with a smaller change (the C code of the tracker and the lead it introduces) and better behaviour on balanced sags; the dq-PLL stays the reference for its simplicity and its exact response to balanced events, but it cannot be used through an asymmetric fault without the gain scheduling, and with it the recovery is the worst of the three.



Table 6: The three PLLs at a glance (numbers of Tables 4 and 5; the DDSRF-PLL in its improved form).

	dq-PLL	FHT-PLL	DDSRF-PLL (improved)
negative sequence (type D dip)	33.6° ripple, 8.6° with the fault gains	1.4° ripple (leak of the calculator)	0.05° ripple, exact steady state
transient at the fault start / end	4.9° / 16° (scheduled)	1.8° / 1.9°	2.5° / 1.3°
phase jump 30°, settling	35 ms	37 ms	107 ms
frequency step 1 Hz	0.8°, no bias	1.6°, bias 0.8° per hertz	3.4°, no bias
balanced sag to 0.3 pu	none	8°	12°
harmonics 5/7/11/13	0.4°	0.4°	0.03°
DC offset 2 %	1.3°	0.9°	0.5°
steady-state offset	0	4.15° (C code)	1.96° (pre-filters)
what to change	release of the scheduled gain, k_i with k_p^2	tracker code (innovation, states, δ), adaptive ω_0	wiring, integrator init and clamp, PI gains

8 Files and how to reproduce

- `foundations_of_power_electronics/three_phase_dqPLLs_analysis` of [1]: the model `frt_dqpll.slx`, its init `init_model.m` (struct `param`, design check `pll_design`) and the new `plotting_results_plls.m`: after a run (or on a saved `sim_results_01.mat`) it runs the six variants of Table 4 on the logged voltage in MATLAB, checks the FHT-PLL branch against the log, prints the table and draws four figures. It reads the gains from `param.afe.ctrl`, the period from `param.glb_time.ts_afe` and the fault window from `param.frt.data`.
- `py/pll_models.py` (this folder): the three loops and their variants in numpy, with the options used in this note (gains, initial frequency, clamp, pre-filters on/off, τ, DC branch, normalization); `py/make_figures.py`: every figure and number of the note, run with the path of the logged `.mat` file (80 MB, not in the repository: `save sim_results_01.mat` after a run of the model with `param.sim.simlength` = 2.25 s) or without it for the analytic and synthetic parts only. The numbers are in `figures/three_phase_plls_numbers.txt`.
- The library [2]:
  - `user_defined_functions/ccaller/src/first_harmonic_tracker.c`: the tracker;
  - `setup_files/objs/ctrl_afe_setup.m`: gains, observer, trackers, τ;
  - `masked_models/filters/discrete_time_low_pass_filter.slx`: the low-pass filter



of the DDSRF.

References

[1] pwr-control, *modelization-and-control*, repository, folder `foundations_of_power_electronics/three_phase_dqPLLs_analysis` (model `frt_dqpll.slx`, README with the design check of November 2026). https://github.com/pwr-control/modelization-and-control

[2] pwr-control, *library*, repository: C callers (`first_harmonic_tracker.c`, `dqpll_grid`), setup classes (`ctrl_afe_setup`), masked blocks (`discrete_time_low_pass_filter`, `afe_uvw_ctrl`). https://github.com/pwr-control/library

[3] D. Bagnara, *AFE control in the uvw frame: the five MATLAB Function blocks of afe_uvw_ctrl*, pwr-control technical note, rev. 01, November 2026, `repos_doc/afe_uvw_ctrl` of *repositories-documentation* (the blocks that read the angle and the frequency of these PLLs).

[4] D. Bagnara, *Single Phase Locked Loop: structures, transfer functions and laboratory implementation*, pwr-control technical note, May 2021, `repos_doc/single_phase_lock_loop` of *repositories-documentation*.

[5] V. Kaura and V. Blasko, "Operation of a phase locked loop system under distorted utility conditions," *IEEE Transactions on Industry Applications*, vol. 33, no. 1, pp. 58–63, 1997.

[6] S.-K. Chung, "A phase tracking system for three phase utility interface inverters," *IEEE Transactions on Power Electronics*, vol. 15, no. 3, pp. 431–438, 2000.

[7] P. Rodríguez, J. Pou, J. Bergas, J. I. Candela, R. P. Burgos and D. Boroyevich, "Decoupled double synchronous reference frame PLL for power converters control," *IEEE Transactions on Power Electronics*, vol. 22, no. 2, pp. 584–592, 2007.

[8] R. Teodorescu, M. Liserre and P. Rodríguez, *Grid Converters for Photovoltaic and Wind Power Systems*, Wiley, 2011, chapter 8 (grid synchronization in three-phase power converters).

[9] VDE-AR-N 4110:2018-11, *Technical requirements for the connection and operation of customer installations to the medium voltage network (TAR medium voltage)*, VDE FNN, 2018.